



Solution Guide: Permissions

Start by inspecting the file permissions on each of the files listed, and determine if they are already set correctly or if you need to change the permissions.

- Run: `ls -l <file1> <file2> <file3>`
- 1. Set permissions 600 on `/etc/shadow` (rw for root only).
 - Running `ls -l /etc/shadow` indicates that the permissions are set to 640.
 - Run: `sudo chmod 600 /etc/shadow`
- 2. Set permissions 600 on `/etc/gshadow` (rw for root only).
 - Running `ls -l /etc/gshadow` indicates that the permissions are set to 640.
 - Run: `sudo chmod 600 /etc/gshadow`
- 3. Set permissions 644 on `/etc/group` (rw for root and r for all others).
 - Running `ls -l /etc/group` indicates that the permissions are already set to 644.
- 4. Set permissions 644 on `/etc/passwd` (rw for root and r for all others).
 - Running `ls -l /etc/passwd` indicates that the permissions are already set to 644.

Bonus

- 5. Verify all accounts have passwords.
 - Running `sudo grep root /etc/shadow` indicates that the root user doesn't have a password.
 - We want to verify that each account has a password hash and not a `!` in the second field of each listing in the `/etc/shadow` file. `!` indicates that there is no password set for that user.
 - Notice that if simply grep for `!`, we can quickly determine if other users have no password, rather than manually inspecting the shadow file.
 - Run: `sudo grep "!" /etc/shadow`
- 6. Verify that no users have a UID of 0 besides `root`. If you find one that does, change its UID to any value greater than 1000.
 - We are examining the third field of each line in the `/etc/passwd` file. Only the root user should have a 0 in this field, and everything else should have a value greater than 1000 if it's a person, and less than 1000 if it's a service user.
 - Running `sudo less /etc/passwd` indicates that the user `adam` also has a UID of 0.
 - Note: A cleaner but trickier solution is to run `grep "x:0" /etc/passwd`. This requires first recognizing that the user ID is preceded by `"x:"`.

- Run `sudo nano /etc/passwd` to change the UID from 0 to something greater than 1000, and that is not in use by another user!
- 7. Add a list of your findings to your research directory.
 - Run `nano ~/research/permissions.txt` to create a document to store your findings, including everything from above.